

GRADUATION PROJECT

Egypt Water impact & Intelligence

Graduation Project Documentation

Project Planning • Literature Review • Requirements Gathering
Team Responsibilities • Project Timeline

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Project Stage: Initial Planning / Repository Setup

GitHub: github.com/nadaxz863/Egypt-Water-impact-Intelligence

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Project Overview

Egypt Water & Economic Intelligence is an integrated data intelligence platform designed to analyze relationships between water resources, agriculture, energy, climate, and economic indicators in Egypt. The project combines SQL, Python, Power BI, Machine Learning, and an AI Agent into one connected platform to explore historical trends, cross-domain relationships, forecasts, and scenario-based estimates.

Project Vision

The platform aims to connect water resources with agriculture, energy, and climate, and explore how these domains relate to economic activity and business indicators. The system will support historical analysis, forecasting where data permits, natural-language questions about project data, and scenario exploration.

Main Data Domains

Domain	Planned indicators
Water	Water availability, renewable water resources, water withdrawal, water use by sector and source, irrigation and drainage, and water stress indicators.
Agriculture	Crop production, harvested area, crop yield, agricultural trends, and irrigation-related indicators.
Energy	Electricity generation and consumption, renewable energy, and solar, wind, and hydro indicators.
Climate	Rainfall, temperature, climate indicators, and drought-related indicators where available.
Economy	GDP, GDP per capita, agriculture's contribution to GDP, inflation, and other relevant indicators.
Business	Enterprise/business indicators, business activity, and constraints where suitable data is available.

Technology Stack

- SQL — database design, transformation, and analytics.
- Python — data cleaning, exploratory data analysis (EDA), and statistical analysis.
- Pandas / NumPy / Matplotlib — data processing, numerical analysis, and visualization.
- Power BI — interactive dashboards and KPI monitoring.
- Machine Learning — forecasting and predictive analysis where data supports it.
- Chatbot Development — a conversational interface that answers users' questions using project data and analytical results.
- Git & GitHub — version control, collaboration, code, and documentation.

1. Project Planning & Management

Project Proposal

The project will build an integrated and explainable data intelligence platform connecting Egypt's water, agriculture, energy, climate, economy, and business indicators. It will combine data engineering, analytics, visualization, machine learning, and an AI Agent rather than creating unrelated dashboards.

Project Plan

The project will proceed through planning, literature review, requirements gathering, system analysis and design, data preparation, SQL development, Python analysis, Power BI development, Machine Learning, AI Agent development, scenario simulation, testing, documentation, and final presentation.

Task Assignment & Roles

The following is the division of work across the six team members.

RAHMA ABDELMUHAYMIN & DUHA KAMAL

Data Cleaning & Data Quality

Responsible for cleaning and preparing the data.

- Receive the raw data from the other team members.
- Review columns, data types, missing values, and duplicates.
- Standardize date formats, units, and column names where needed.
- Resolve data issues and document every change made.
- Save a Cleaned copy of each dataset while keeping the Raw Data unchanged.
- Work with each team member to understand the nature of their data before cleaning it.

Tools: Python, Pandas, and Excel when needed.	**Final Deliverable:** Cleaned Datasets ready to be loaded into SQL.

NADA AYMAN & NADA ELSAYED

SQL & Database

Responsible for building the database.

- Design the database schema and the appropriate tables.
- Define primary keys, foreign keys, and the relationships between tables.
- Create the tables using SQL.
- Load the Cleaned Datasets into the database.
- Write SQL queries to extract the required data and indicators.
- Validate the data after loading, and collaborate with the Data Modeling & Analysis team to prepare tables for analysis.

Tools: SQL Server or PostgreSQL.	**Final Deliverable:** An organized, integrated database that is ready for analysis and for connecting to Power BI.

NIERA ELSHAMY & AYA KHALED

Data Modeling & Analysis

Responsible for analyzing the data and extracting insights.

- Define the appropriate relationships between water, agriculture, energy, climate, and economy data.
- Review the data model together with the SQL & Database team.
- Perform Exploratory Data Analysis (EDA).
- Calculate the KPIs and key indicators.
- Analyze trends and changes over the years.
- Study relationships between variables, keeping in mind that correlation does not necessarily imply causation.
- Prepare the data and features required for Machine Learning together with the Machine Learning & AI Agent team.

Tools: Python, Pandas, Matplotlib, and SQL.	**Final Deliverable:** Documented analyses, KPIs, and insights, plus data ready for the Dashboard and Machine Learning.

NADA AYMAN, NIERA ELSHAMY, DUHA KAMAL & RAHMA ABDELMUHAYMIN

Power BI & Dashboard

Responsible for designing the interactive report.

- Connect Power BI to the SQL database.
- Prepare the table relationships and Measures using DAX.
- Design one unified dashboard for the project, not five separate dashboards.
- Create pages for water, agriculture, energy, climate, and economy.
- Design an Executive Overview that shows the key indicators of the project.
- Add filters, slicers, and suitable charts.
- Review the numbers with the Data Modeling & Analysis team to confirm they are correct.

Tools: Power BI, DAX, and SQL.	**Final Deliverable:** An interactive Power BI report with a unified design that presents the project results.

NADA ELSAYED & AYA KHALED

Machine Learning & AI Agent

Responsible for the intelligent components and the final integration.

- Define a forecasting problem that suits the available data.
- Prepare, train, and evaluate the ML model using suitable metrics.
- Develop an AI Agent that can answer users' questions about the project data.
- Connect the Agent to trusted data sources or analytical results.
- Test the correctness of the answers and the forecasting results.
- Coordinate with the SQL & Database team for access to SQL data, and with the Data Modeling & Analysis team for features and analytical results.
- Join the team in integration testing of the project components.

Tools: Python, suitable ML libraries, and AI Agent building tools according to the design.	**Final Deliverable:** A suitable forecasting model and an integrated AI Agent, depending on the project scope and the available time.

Shared Responsibilities

- Rahma Abdelmuhaymin and Duha Kamal work with every team member to understand each dataset before cleaning it, and hand over the Cleaned Datasets for loading into SQL.
- Nada Ayman and Nada Elsayed collaborate with Niera Elshamy and Aya Khaled on the data model and on preparing the tables for analysis.
- Niera Elshamy and Aya Khaled prepare the features required for Machine Learning together with the Machine Learning & AI Agent team, and review the dashboard numbers with the Power BI team.
- All six members contribute to requirements review, result validation, integration testing, documentation, and final presentation.

Risk Assessment & Mitigation Plan

Risk	Mitigation
Missing or incomplete data	Use documented public sources, assess coverage early, and clearly report gaps.
Inconsistent units or time periods	Standardize units and dates; document transformations before integration.
Incompatible datasets or unclear join keys	Define integration keys explicitly and avoid unsupported joins.
Limited data for forecasting	Choose ML targets only after assessing historical coverage and data quality.
Misinterpreting correlation as causation	Describe relationships carefully and state analytical limitations.
AI Agent producing unsupported answers	Ground responses in project data and analytical outputs; test against known results.
Schedule or integration delays	Track milestones weekly, prioritize a working end-to-end version, and keep optional features scoped.

KPIs (Key Performance Indicators)

The final KPI list will depend on source availability, coverage, compatible units, and analytical suitability.

- **Water:** water availability, renewable water resources, water withdrawal, water use by sector/source, and water stress.
- **Agriculture:** crop production, harvested area, crop yield, and agricultural trends.
- **Energy:** electricity generation and consumption, and renewable energy indicators.
- **Climate:** temperature, rainfall, and drought-related indicators where available.
- **Economy:** GDP, GDP per capita, agriculture's contribution to GDP, and inflation.
- **Model evaluation:** suitable error metrics for forecasting models, selected according to the target and validation design.
- **AI Agent:** answer correctness against project data, traceability to analytical outputs, and successful handling of supported questions.
- **Dashboard performance:** report pages, filters, and slicers respond quickly enough for smooth interactive use.
- **Usage and adoption:** how often the dashboard and the AI Agent are used by their intended users, and whether users find them useful.
- **Data quality:** coverage and completeness of the datasets, with missing values and duplicates handled and documented.

2. Literature Review

Existing Research & Related Projects

Review research and public datasets concerning Egypt's water resources, agricultural production, energy, climate, and economic indicators. Initial datasets include AQUASTAT for water resources and water-use indicators, and FAOSTAT for crop production, harvested area, and yield. Additional sources will be selected based on reliability, historical coverage, data quality, compatible time periods, clear join keys, and suitability for analysis.

Suggested Improvements

- Add reliable datasets with compatible historical coverage across the project domains.
- Improve standardization and documentation of data transformations.
- Expand cross-domain analysis where data supports valid comparisons.
- Evaluate forecasting models and report their limitations transparently.
- Ground the AI Agent in structured project data and analytical outputs.
- Add a scenario simulator with explicit assumptions and clearly labeled model-based estimates.

Final Grading Criteria

Use the grading distribution supplied by the DEPI program. The categories provided are reproduced below; these weights total 100 points.

Criterion	Weight
Idea & Objectives	10/100
System Analysis & Requirements / Design	15/100
Algorithms / Methods / Techniques	20/100
Implementation & Coding / Results	20/100
Testing & Documentation	10/100
Innovation & Added Value	10/100
Presentation & Demo	10/100
Final Submission & Delivery	5/100
Total	**100/100**

3. Requirements Gathering

Stakeholder Analysis

Potential users include researchers, policymakers, resource planners, and economic analysts interested in exploring the relationships between water resources, agriculture, energy, climate, economy, and business indicators. These are potential stakeholder groups, not confirmed project partners.

Stakeholder Needs

Stakeholder	Needs
Researchers	Reliable, documented datasets with traceable sources, and the ability to explore historical trends and relationships across domains.
Policymakers	Clear summaries of water, agriculture, energy, climate, and economic trends, and scenario-based estimates with clearly stated assumptions.
Resource planners	Insight into water availability, withdrawal, use by sector and source, and water stress, supported by forecasts where data permits.
Economic analysts	Economic indicators such as GDP and inflation connected to water, agriculture, energy, and climate indicators.

User Stories & Use Cases

Use Cases

- Explore historical trends in Egypt's water withdrawal and availability.
- Compare crop production, harvested area, and yield over time.
- Explore electricity generation, consumption, and renewable energy trends.
- Examine temperature, rainfall, and other available climate indicators.
- Investigate relationships across water, agriculture, energy, climate, and economic indicators.
- Ask natural-language questions about project data through the AI Agent.
- Explore a hypothetical scenario, such as a 10% decrease in water availability, and inspect model-based estimates where suitable models exist.

User Stories

- As a researcher, I want to explore historical trends in Egypt's water withdrawal and availability, so that I can understand how water resources have changed over time.
- As a resource planner, I want to compare crop production, harvested area, and yield over time, so that I can see how agricultural output relates to water use.
- As a policymaker, I want to explore electricity generation, consumption, and renewable energy trends, so that I can understand how energy relates to water and the economy.
- As a researcher, I want to examine temperature, rainfall, and drought-related indicators, so that I can place water and agriculture trends in their climate context.
- As an economic analyst, I want to investigate relationships across water, agriculture, energy, climate, and economic indicators, so that I can see how these domains are connected.
- As a user, I want to ask natural-language questions about project data through the AI Agent, so that I can get answers grounded in the project data without writing queries.

- As a policymaker, I want to explore a hypothetical scenario, such as a 10% decrease in water availability, so that I can inspect model-based estimates together with their stated assumptions.

Functional Requirements

- Collect and integrate reliable, documented datasets.
- Preserve raw data separately from cleaned data.
- Clean, validate, standardize, and document transformations using Python and Pandas.
- Design a SQL database with appropriate fact tables, dimension tables, and analytical queries.
- Perform exploratory data analysis, statistical analysis, trend analysis, and cross-domain analysis.
- Build one connected Power BI report with Executive Overview, Water, Agriculture, Energy, Climate, and Economy & Business pages.
- Evaluate Machine Learning models for selected forecasts, depending on data quality and historical coverage.
- Develop an AI Agent that answers natural-language questions using structured project data and analytical outputs.
- Develop a scenario simulator for exploring model-based estimates under stated assumptions.
- Provide technical documentation, a final report, presentation, and demonstration.

Non-functional Requirements

- Data reliability and validation.
- Documented, reproducible transformations.
- Consistent units and compatible time periods.
- Clearly defined data integration keys.
- Maintainable and version-controlled code.
- Understandable and usable visualizations.
- Transparent assumptions and model limitations.
- AI responses traceable to available project data and analytical outputs.
- Performance: dashboards, queries, and AI Agent responses are fast enough for interactive use.
- Security: access to the database, accounts, and project files is limited to authorized team members, and credentials are kept out of the GitHub repository.

Project Architecture

Data Sources → Data Collection → Data Cleaning & Validation → SQL Database / Data Model → Python EDA & Analytics → Power BI Dashboards → Machine Learning → AI Agent

Planned Power BI Report

Report Page	Planned Content
Executive Overview	Summary of available water, agriculture, energy, economy, and climate indicators.
Water	Water resources, water withdrawal by sector/source, irrigation, and related indicators.
Agriculture	Crop production, harvested area, yield, and long-term crop trends.
Energy	Electricity generation, consumption, and renewable energy indicators.
Climate	Rainfall, temperature, and available climate trends.
Economy & Business	GDP, agriculture's contribution to GDP, inflation, and business indicators where data is available.

Machine Learning

After the analytical layer is established, suitable models will be evaluated for forecasting selected indicators, which may include water availability or demand, crop production, electricity demand, and selected economic indicators. Final targets and models depend on data quality, historical coverage, and validation results.

AI Agent

The AI Agent will provide a natural-language interface to supported project data and analytical outputs. Example questions include: "What are the main trends in Egypt's water withdrawal?", "Which crops have the highest production?", and "How has crop yield changed over time?" The agent should not present unsupported claims as findings.

Scenario Simulator

A later-stage component may allow users to explore scenarios such as a hypothetical 10% decrease in water availability. Outputs must be based on available models and clearly stated assumptions. They are model-based estimates, not proof of causation or guaranteed predictions.

Initial Datasets & Data Principles

- AQUASTAT — initial source for water resources and water-use indicators for Egypt.
- FAOSTAT — initial source for crop production, harvested area, and yield for Egypt.

- Additional economy, energy, climate, and business sources will be selected after evaluating reliability, historical coverage, data quality, compatible time periods, join keys, and analytical suitability.

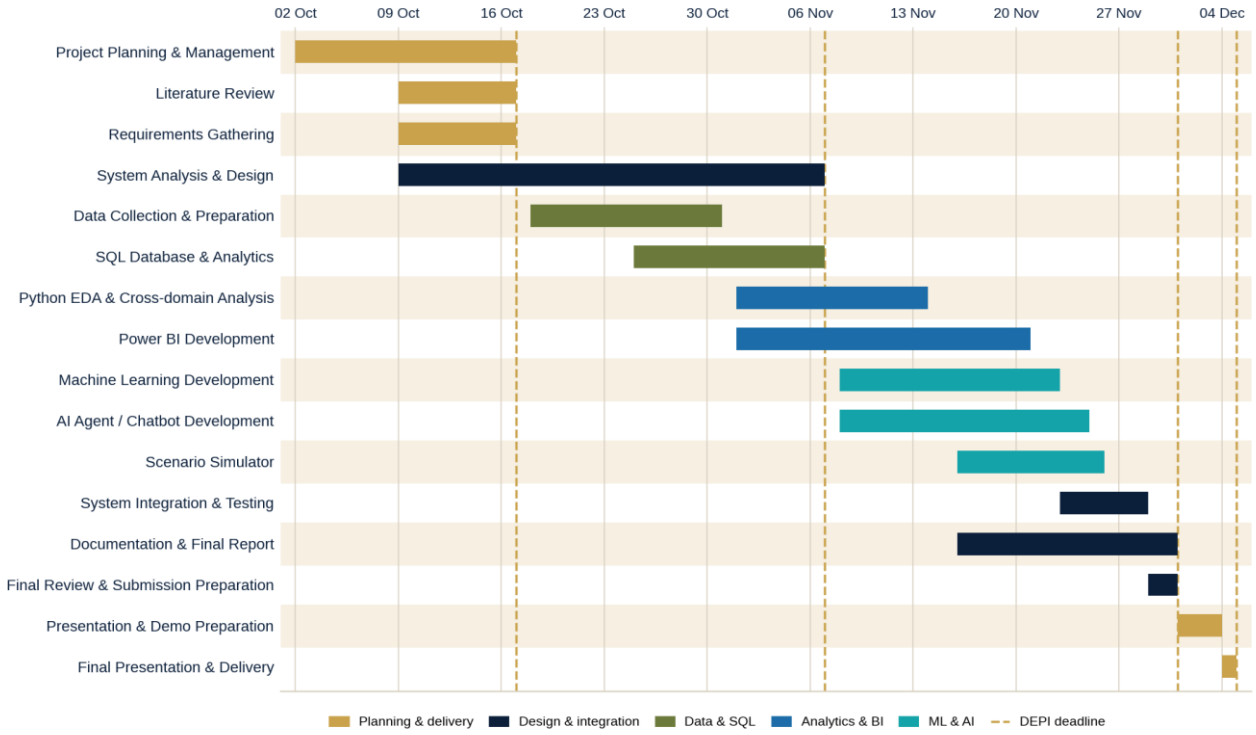
Data & Analysis Principles

- Use reliable and documented sources.
- Preserve original/raw data separately from cleaned data.
- Document data transformations.
- Avoid mixing incompatible units or time periods.
- Align datasets using clearly defined keys.
- Distinguish correlation from causation.
- Clearly label forecasts and scenario estimates as model outputs.

4. Project Plan (Gantt Chart)

The milestones below use the dates supplied in the DEPI schedule. Internal task windows are proposed planning dates and should be adjusted by the team as work progresses.

Gantt Chart



Task Schedule

Task / Phase	Start	End	Responsible	Deliverable
Project Planning & Management	2 Oct 2026	16 Oct 2026	All members	Proposal, plan, roles, risks
Literature Review	9 Oct 2026	16 Oct 2026	Nada Ayman, Nada Elsayed, Niera, Aya	Related research and data-source review
Requirements Gathering	9 Oct 2026	16 Oct 2026	All members	Requirements, stakeholders, use cases
System Analysis & Design	9 Oct 2026	6 Nov 2026	Nada Ayman, Nada Elsayed, Niera, Aya	Architecture, database and report design
Data Collection & Preparation	18 Oct 2026	30 Oct 2026	All members; Rahma and Duha lead cleaning	Collected, cleaned, validated datasets
SQL Database & Analytics	25 Oct 2026	6 Nov 2026	Nada Ayman, Nada Elsayed	Database, queries and KPIs
Python EDA & Cross-domain Analysis	1 Nov 2026	13 Nov 2026	Niera, Aya	Analysis, trends and prepared features
Power BI Development	1 Nov 2026	20 Nov 2026	Nada Ayman, Niera, Duha, Rahma	Connected report and analytical pages
Machine Learning Development	8 Nov 2026	22 Nov 2026	Nada Elsayed, Aya	Trained/evaluated models
AI Agent / Chatbot Development	8 Nov 2026	24 Nov 2026	Nada Elsayed, Aya	Data-aware chatbot
Scenario Simulator	16 Nov 2026	25 Nov 2026	Nada Elsayed, Aya	Scenario estimates and visualization
System Integration & Testing	23 Nov 2026	28 Nov 2026	All members	Integrated and tested system
Documentation & Final Report	16 Nov 2026	30 Nov 2026	All members	Technical docs and final report
Final Review & Submission Preparation	29 Nov 2026	30 Nov 2026	All members	Final code and submission package
Presentation & Demo Preparation	1 Dec 2026	3 Dec 2026	All members	Slides, demo, rehearsal
Final Presentation & Delivery	4 Dec 2026	4 Dec 2026	All members	Final presentation and delivery

Official DEPI Milestones

Milestone	DEPI Deadline
Project Planning & Management	16 Oct 2026
Literature Review	16 Oct 2026
Requirements Gathering	16 Oct 2026
System Analysis & Design	6 Nov 2026
Implementation (Source Code & Execution)	30 Nov 2026
Final Presentation, Testing & Reports	4 Dec 2026

> **Scheduling note:** As of 18 October 2026, the planning, literature review, and requirements deadlines have passed. If any deliverables remain incomplete, prioritize them immediately while beginning data preparation. Internal dates are proposed working dates; the official DEPI deadlines above take precedence.

5. DEPI Grading Rubric

Criterion	Weight
Idea & Objectives	10/100
System Analysis & Requirements / Design	15/100
Algorithms / Methods / Techniques	20/100
Implementation & Coding / Results	20/100
Testing & Documentation	10/100
Innovation & Added Value	10/100
Presentation & Demo	10/100
Final Submission & Delivery	5/100
TOTAL	**100/100**

Team & Current Status

Team Members:

- Aya Khaled
- Duha Kamal
- Rahma Abdelmuhaymin
- Nada Ayman
- Nada Elsayed
- Niera Elshamy

Supervisor: Eng / Maged Magdy

Current Stage: Initial Planning / Repository Setup. The repository contains the initial project architecture and documentation. Data collection, SQL modeling, analytics, dashboards, Machine Learning, AI Agent development, and scenario simulation will be added progressively.

GitHub Repository: github.com/nadaxz863/Egypt-Water-impact-Intelligence

Project Goal

> *Build an integrated and explainable data intelligence platform that connects Egypt's water, agriculture, energy, climate, economy, and business indicators to support analysis, forecasting, and scenario exploration.*